SAMPLE

# Cosmos Unveiled: The Epoch of Reionization

When Light Broke Through

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## Introduction

Between the afterglow of creation and the starry sky we know, the universe spent its first few hundred million years in a darkness that almost no light relieved. This book is the account of that darkness and of its ending: the epoch of reionization, the period when the first generation of stars and galaxies tore the cosmic gas apart, atom by atom, and made the universe transparent to the light that now crosses it. The story is one of the youngest subjects in astronomy, and one of the most demanding, because its evidence is scattered across quasar spectra, the microwave background's polarization, and the faintest galaxies the telescopes have ever seen. The reader is invited into a detective story whose crime scene is the entire cosmos.

A word on the book's place in the wider series belongs here. The reader who has traveled from the plasma of the first moments, through the formation of the light elements and the release of the background radiation, arrives at this volume at the last of the universe's global transitions: the story of what happened when the primordial gas met the first light. The chapters that follow assume little and derive much, and the reader needs no more than patience with a subject whose evidence is scattered and whose certainty comes in brackets rather than declarations. The reward for the patience is the picture itself: a young universe, dark and simple, ending its childhood in a single spectacular era, the account this book exists to give.

The chapter's plan is simple. First, the setting: the dark ages, the era of neutral hydrogen that followed the initial cooling, when no star had yet formed and the universe was a nearly featureless fog. Second, the mechanism: the ultraviolet photons that escaped the first stars and galaxies, and the ionization physics that let them carve bubbles of plasma into the surrounding gas. Third, the evidence: the absorption spectra of distant quasars, the microwave background's faint scattering signature, and the sudden drop in distant Lyman-alpha emitters. Fourth, the frontier: the 21-centimeter signal, the radio signature of neutral hydrogen itself, which the coming decade's instruments hope to detect. The conclusion places reionization in the arc of cosmic history as the last global transition the universe underwent, the moment the cosmic web acquired its modern character.

An image is worth setting out before the chapters begin, because the subject's name gives almost nothing away. For a few hundred million years after the first stars, the universe was a mist being burned off: hydrogen atoms, neutral since the afterglow era, floated in a dark between the young galaxies, and starlight had to eat its way through them, bubble by bubble, until the last fog cleared and the cosmos became transparent to the light it still is. This book is an account of the last phase change of the universe, and the chapters that follow are the detective case: how we know it happened, who did it, and when, reconstructed from the faintest evidence in all of astronomy.

A second framing fact sets the subject's difficulty, and it deserves to be held with respect. The epoch left no single picture: its light, stretched by expansion into the infrared and the radio, arrives as signatures rather than images, a wobble in the afterglow, a gap in a beacon's spectrum, a whisper in a wavelength no eye sees, and each chapter of this book is the account of one signature and the instrument built to read it. Reionization is consequently the least observed era the reader will ever study, and the most triangulated, a whole history assembled from shadows, which is exactly why the subject's practitioners treat its every measurement as a prize.

## Chapter 1: The Setting - A Universe of Neutral Hydrogen

The book's evidence geography deserves one last map before the journey. The quasar spectra of Chapter 4 were gathered on the largest telescopes of the ground, and the polarization maps of Chapter 5 from the great satellites' scans; the emission-line surveys of Chapter 6 come from the deepest imaging and the patient spectroscopy; the 21-centimeter limits of Chapter 7 from the radio arrays on three continents; the galaxy audits of Chapter 9 from the newest flagship's cold mirror; and the simulations of Chapter 10 from some of the largest computers ever built. It pays to see the map's lesson before the chapters draw it in detail: no observatory sees this epoch, and every kind of instrument contributed its fragment, a subject assembled from the whole toolkit of a whole science.

The story begins where the microwave background ends. At an age of roughly 380,000 years, the universe cooled below the temperature at which electrons and protons can stay bound, and the plasma combined into neutral hydrogen. The opaque fog cleared, the radiation streamed free, and that freed radiation, stretched by a factor of eleven hundred, is the microwave background we observe today. The moment of combination is the starting line of this book: from that instant, the universe consisted of neutral hydrogen, a little helium, traces of lithium, and the dark matter's invisible scaffolding, with no heavier elements and no source of light anywhere. Hold close the

scene's austerity. The cosmos at combination was smooth, dark, and simple, and everything since has been elaboration.

The relative timescales deserve a paragraph before the darkness deepens, since the reader's intuition needs calibrating. The universe's plasma era lasted a few hundred thousand years; the dark ages that followed, a few hundred million; the reionization era, a few hundred million more; and the universe's whole subsequent history, the galaxies' maturity, the stars' long generations, and the reader's own era, the remaining thirteen billion. Bear in mind the proportions' oddness: the darkness the book is about was not a brief prelude but a stretch comparable to the epoch that ended it, and the universe spent as long unlit as it did clearing, the three eras of the early cosmos of nearly equal length, a symmetry the casual telling of cosmic history rarely conveys.

The dark ages that followed deserve their name. The gas, released from the radiation's grip, began to fall into the dark matter's halos, whose seeds had been laid by the inflationary era's quantum ripples. There was structure in the dark: the density fluctuations measured in the microwave background, the parts-in-a-hundred-thousand wrinkles that gravity would slowly amplify. But for a hundred million years and more, nothing in the universe shone. The only glow was the fading afterglow of the background itself, reddening as the expansion stretched it, and the feeble warmth of the primordial gas's own heat, trapped and leaking. A note on the dark ages' duration and its drift into the light deserves a paragraph. The era's length, from combination to the first star formation, was a few hundred million years, the time the smallest halos needed to gather and cool their gas; the transition was not a moment but a ramp, the first stars igniting here and there over tens of millions of years, the earliest sparks appearing while most of the volume still lay in the original darkness. Keep hold of the era's character as gradual: the dark ages ended locally and repeatedly, in thousands of separate places, and the global epoch this book describes, the reionization, is simply the sum of all those local endings, the fog's clearing as the accumulation of a million small dawns.

Refuse the picture of a black starfield: there were no stars, no galaxies, no black holes, and the universe's entire luminosity was the relic of its birth, declining toward an infrared murmur.

The microwave background's own record of the seeds deserves a paragraph here, because the whole later story is the working-out of its numbers. The fluctuations, mapped in exquisite detail by the satellite missions, are parts in a hundred thousand: the density's fractional variations, frozen into the radiation at combination, with a characteristic scale that the acoustic waves of the plasma era impressed. The halo population that later formed is the direct descendant: the peaks of the initial field becoming the first minihalos, the larger peaks the clusters' seeds, and the statistics of the later bubbles' placement the statistics of the earlier wrinkles, transformed by gravity. Carry forward the continuity's meaning: the reionization era's patchiness was not arbitrary but inherited, the light's geography following the darkness's geography, and the whole account of this book is, in a sense, the microwave background's closing chapter, the wrinkles' final consequence in light.

The temperature history matters for what follows. As the universe expanded, the gas cooled in step with the stretching of the radiation, both falling together until the gas decoupled and then cooled faster, its temperature tracking the expansion directly. By a few hundred million years, the primordial gas had chilled toward the tens of kelvins, cold enough that its hydrogen should have emitted a faint radio line, and it is precisely this line, the 21-centimeter radiation of neutral hydrogen, that the newest observatories are built to find. Record the fact: the dark ages are dark at optical wavelengths but murmuring at radio ones, and the signal's strength depends on the gas being colder than the radiation behind it.

The minihalo earns a paragraph to itself, as the first object of the story. The smallest gravitationally bound structures of the hierarchy, masses of a few hundred thousand suns, were the sites where the gas first collected: too small to hold atomic hydrogen's heat, they required the hydrogen molecule, formed in the sparse gas by the rare two-step pathways, whose rotational lines cooled the gas by tens of kelvins and let it collapse to the densities of stellar birth. The minihalo's fate is brief: the first star that forms inside it heats the remaining gas, drives it out, and ends the halo's career as a nursery, and the whole cycle, from first collapse to stellar ignition, is the dark ages' closing act. Mark the scale's humbling smallness: the universe's first lights ignited in structures a million times lighter than our galaxy, the humblest addresses in the cosmos, and the entire epoch this book recounts began in them.

The formation of the first objects closes the dark ages. In the halos where the dark matter's density peaked, the gas gathered into the first bound clumps, and when a halo's mass passed a threshold set by the gas's temperature, hydrogen molecules formed, cooled the gas further, and let it collapse. The very first stars, the so-called Population III, were born from pristine gas with no metals to help it cool, and the theory predicts they

were massive, hot, and short-lived, their surfaces blazing with the ultraviolet that would begin the reionization story. Treat gently the book's central irony gently: the darkness was ended by objects that no telescope has yet seen directly, and the epoch's whole drama rests on stars whose existence we infer from their consequences.

The setting file adds the arithmetic of the fog's raw material, because the numbers make the darkness tangible. The neutral hydrogen that filled the dark ages amounted to roughly one atom for every cubic meter of space, a better vacuum than any laboratory on the Earth has ever made, yet the volume was so vast that the total mass outweighed all the stars that would ever form; the book's subject is accordingly the clearing of a medium thinner than nothing and heavier than everything, and the arithmetic in this chapter disciplines the imagination as it goes: the fog was not a mist one could wade through but an emptiness that absorbed light anyway, one whispering atom at a time.

The setting file also adds the sound note, because the dark ages were not silent. The young universe rang with pressure waves through its fog, the ripples the afterglow's maps show as hot and cold spots, and the dark ages inherited their pattern: the first structures condensed along the wave-crests, the first lights ignited where the fog had been compressed, and the epoch's patchiness that the later chapters chase is the standing echo of that ancient sound. The addition on the note is the setting's last word before the first light: the darkness had a shape before it had an end, the shape was written by sound, and the clearing of the fog merely revealed the melody that had been in it all along.